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SYRINGIC ALDEHYDE

[Syringaldehyde]



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1. Procedure

A well-stirred (Note 1) and (Note 2) mixture of 740 ml. of [glycerol](#) and 154 g. (1 mole) of [pyrogallol-1,3-dimethyl ether](#) and 154 g. (1.1 mole) of [hexamethylenetetramine](#) is placed in a 1-liter, three-necked round-bottomed flask fitted with a thermometer and a condenser for downward flow. The mixture is heated in an oil bath to 170° . This temperature is maintained for 30 minutes and then allowed to fall to 150° . The [boric acid](#) is added as rapidly as possible through the neck holding the thermometer. The temperature is immediately started but is slowed down when the temperature begins to rise. The reaction is watched and controlled very carefully when this temperature is reached, since it is very exothermic. The temperature is maintained at 150 – 160° for 5 to 9 minutes (Note 7). At the end of this reaction time the mixture is cooled to 110° as rapidly as possible (Note 6) and (Note 8), and a previously prepared solution of 184 ml. of concentrated [sulfuric acid](#) in 620 ml. of water is added to the reaction mixture. After being stirred for 1 hour, the mixture is cooled to 25° in an ice bath. The [boric acid](#), which separates from the solution, is removed by filtration (Note 9) and washed free of mother liquor with 400 ml. of water. The filtrate and washings are combined and extracted with three 500-ml. portions of [chloroform](#) (Note 10), (Note 11), and (Note 12).

The [chloroform](#) solution is then extracted with a filtered solution of 180 g. of [sodium bisulfite](#) in 720 ml. of water (Note 13) by stirring rapidly with a Hershberg stirrer for 1 hour. The separated bisulfite solution is washed twice with [chloroform](#), filtered, and acidified in a hood with a solution of 55 ml. of concentrated [sulfuric acid](#) in 55 ml. of water. After careful heating on a steam bath for a short time, air is bubbled through the hot solution until all the [sulfur dioxide](#) has been expelled. The product, which separates as a mixture of crystals and oil, readily solidifies upon cooling (Note 14). The [syringic aldehyde](#) is collected by filtration, washed with cold water, and dried in an oven at 40° to give 62.5–66 g. of light-tan material, melting at 110.5 – 111° , which still contains a small amount of foreign material that does not melt at 300° . Recrystallization of the crude product from aqueous [methanol](#) using 30 ml. of water and 3 ml. of [methanol](#) for each 10 g. of aldehyde gives 56–59 g. (31–32%) of product melting clear at 111 – 112° (uncor.). A second extraction of the [chloroform](#) solution with a filtered solution of 60 g. of [sodium bisulfite](#) in 240 ml. of water gives an additional 3–4 g. of product.

2. Notes

- The use of a Hershberg² stirrer is recommended.
- It is desirable to conduct this preparation in a hood because of the large volume of [ammonia](#) liberated in the second step.
- Eastman Kodak Company white label grade [pyrogallol-1,3-dimethyl ether](#) was used. A larger excess of [hexamethylenetetramine](#) in a small trial run did not improve the yield.
- The reaction mixture darkens rapidly, and there is a vigorous evolution of [ammonia](#).
- The temperature usually rises to 160° within 5 minutes, and cooling is necessary.
- Cooling is accomplished by playing a stream of cold water over the outside of the flask.
- There is undoubtedly some leeway in these conditions. The same yields were obtained when the temperature was maintained at 150 – 160° for periods of 5 to 9 minutes. Longer reaction times, without rapid cooling after the heating period, lowered the yield. Reaction times of 15 minutes, 30 minutes, and 1 hour gave yields of 20.8%, 10.0%, and 6.5%, respectively. Liggett and Diehl,³ after having run a large number of other Duff reactions, have come to the conclusion that the temperature may vary between 145° and 175° without detriment to the yield.
- About 3 to 5 minutes is required for cooling.
- If not removed, the [boric acid](#) makes extraction of the product impossible or very difficult. Since the [boric acid](#) is finely divided, filtration is extremely slow unless large Büchner funnels, preferably with large holes, are employed. The checkers avoided the difficulty by using a filter cloth at this point.
- The product cannot be isolated by steam distillation of the reaction mixture.
- [Syringic aldehyde](#) is much more soluble in [chloroform](#) than in [ether](#). Extraction is essentially complete since a fourth extraction gave only a 0.3–0.7% increase in yield.
- If the aldehyde is isolated directly by concentration of the [chloroform](#) solution, the color is darker, and the melting point and yield are lower.
- This represents a large excess of [sodium bisulfite](#), but smaller amounts remove a smaller percentage of the [syringic aldehyde](#) from the [chloroform](#) solution. When larger amounts of bisulfite are employed, extraction of the product is still incomplete.
- The product should be cooled to just 15° and filtered immediately. Longer and further cooling causes [sodium sulfate](#) to crystallize from the mixture. Very little product remains in the filtrate.

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The paragraphs above were added in September, 2014. The statements above do not supersede any specific hazard caution notes and safety instructions included in the procedure.

3. Discussion

This procedure is a modification of the method described by Manske and co-workers.⁴ [Syringic aldehyde](#) has also been obtained by numerous other procedures from [pyrogallol-1,3-dimethyl ether](#)^{5,6,7,8} from gallic acid,^{9,10,11} and from [vanillin](#).¹²

References and Notes

1. Eastman Kodak Company, Rochester, New York.
2. [Org. Syntheses Coll. Vol. 2, 117 \(1943\)](#).
3. Liggett and Diehl, *Proc. Iowa Acad. Sci.*, **52**, 191 (1945).
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Appendix

Chemical Abstracts Nomenclature (Collective Index Number); (Registry Number)

sulfuric acid (7664-93-9)

ammonia (7664-41-7)

methanol (67-56-1)

ether (60-29-7)

chloroform (67-66-3)

glycerol (56-81-5)

sulfur dioxide (7446-09-5)

sodium sulfate (7757-82-6)

sodium bisulfite (7631-90-5)

hexamethylenetetramine (100-97-0)

boric acid (10043-35-3)

vanillin (121-33-5)

Syringic aldehyde,
Syringaldehyde (134-96-3)

pyrogallol-1,3-dimethyl ether

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